

Information Retrieval
Programming Assignment I
Preprocessing, Indexing, Boolean Search
Submission deadline: August 31.

General Guidelines: The project is to be done in a group. ONLY ONE MEMBER PER GROUP SHOULD SUBMIT THE ASSIGNMENT IN MOODLE. Plagiarism will be penalised. Marks will be awarded depending on completeness and correctness. You can write your programs either in C/C++/Java/python language. You should not use any information retrieval library and write the programs from scratch. Basic string processing libraries may however be used. The Porter stemmer program from any source may be used. All your programs and output file names should be prefixed by your group name.

Problem Description:

The goal of this assignment is to write programs to preprocess the text documents in a collection and prepare an index file. The tokenization, normalization, stemming and stop word removal is to be performed before the indexing. The index file will be searched to answer Boolean queries using the Boolean retrieval model.

Text Collection: We will use the Cranfield text collection containing 1400 documents. The documents consist of title, authors, and abstracts of scientific articles. Only the title and the abstracts are to be processed, the author field may be ignored. All the documents in the collection are contained in a single text file **cran.all**. Each document has a tag .I representing the docid, .A, .B representing the authors and their affiliation, .T representing the title, and .W representing the abstract.

Preprocessing:

- i. Tokenization – convert a document into a collection of token strings
- ii. Stemming – convert the tokens to their word stems using the Porter stemmer
- iii. Stop word removal – remove the stop words from the token collection
- iv. Normalization – normalize the tokens

You should write four different functions to implement each of the steps. Input to the tokenization function will be the **cran.all** file. Integrate the functions in a single preprocessing program. The output of the function should be a single text file **groupname_processed.all**. The file should annotate the doc id using the tag .I and the tokens using the tag .S.

You may use the Porter's stemmer program from any source, and a list of English stop words.

Indexing:

Write a function to create a simple index file for the above collection named **groupname_cran.index**. The file should contain every token to be indexed in a separate line, sorted in lexicographical order. The line should contain the postings list containing comma separated *docids* in which the token appears in ascending order. For example, the index file may contain lines of information as –

```
588, 1400
aerodynamic 1,10, 11
experimental 1,7,9,21,27
slipstream 1,532
```

The first two integers in the index file represents the vocabulary size (number of tokens) and the maximum docid that has been indexed.

Boolean Search:

Write programs to perform simple Boolean search involving only two query words and logical connectives AND, OR. Note that the query should also be stemmed and normalized. The program should take as input a Boolean query and return a file containing the list of docid that matches the query using the Boolean retrieval model. To answer the query search should be performed over the index file. Bonus marks for efficient searching algorithms.

Submissions:

Preprocessing, indexing, Boolean search functions, the final index file and the preprocessed text file. The result docid list for some test Boolean queries should also be submitted. Test queries will be provided later. The groups may be asked to demonstrate their outputs. A writeup on the methodology and the program listing including the readme files should compulsorily be submitted. You are encouraged to create a github page for the assignment.